SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1. Ans: (a) The knowledge base contains three diagnostic rules and the following facts about Patient A: **Fever = True, Cough = True, Breathlessness = True**.

Let the following propositions represent the conditions and diagnoses:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **B** = Patient A has Breathlessness
- **R** = Patient A has Rash
- **FL** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia

Rules

Rule 1: If a patient has Fever AND Cough → Possible Flu

$F \wedge C \rightarrow FL$

Rule 2: If a patient has Fever AND Rash → Possible Measles

$F \wedge R \rightarrow M$

Rule 3: If a patient has Cough AND Breathlessness → Possible Pneumonia

$C \wedge B \rightarrow P$

Facts about Patient A

F = True

C = True

B = True

Therefore, the propositional knowledge base is:

$KB = \{F, C, B, (F \wedge C \rightarrow FL), (F \wedge R \rightarrow M), (C \wedge B \rightarrow P)\}$

(b) Forward Chaining is a data-driven inference method. It starts with the known facts and repeatedly applies rules whose conditions are satisfied.

Step 1: Initial Facts

The facts given for Patient A are:

F = True

C = True

B = True

So:

F, C, B

Step 2: Apply Rule 1

Rule 1:

F ^ C → FL

We have:

F = True

and

C = True

Therefore, both conditions are satisfied.

Hence:

FL = True

Derived fact: Patient A has **Possible Flu**.

Step 3: Apply Rule 2

Rule 2:

F ^ R → M

We have:

F = True

But there is **no fact stating that Rash is true**.

Therefore, Rule 2 cannot be fired.

M cannot be derived

Step 4: Apply Rule 3

Rule 3:

C ^ B → P

We have:

C = True

and

B = True

Therefore, both conditions are satisfied.

Hence:

P = True

Derived fact: Patient A has **Possible Pneumonia**.

Forward Chaining Table

Step	Known Facts	Rule Fired	New Fact
1	F, C, B	Initial facts	F, C, B
2	F, C	Rule 1:	**FL**

Step	Known Facts	Rule Fired	New Fact
3	F, no R	Rule 2 cannot fire	—
4	C, B	Rule 3:	**P**

Possible Measles cannot be derived because the fact Rash = True is not provided.

(c) Goal

The required goal is:

P

where **P = Patient A has Possible Pneumonia**.

Backward Chaining starts from the goal and works backward to find rules that can prove the goal.

Step 1: Start with the Goal

P

We search the knowledge base for a rule whose conclusion is .

Rule 3 is:

$C \wedge B \rightarrow P$

Therefore, to prove, we must prove:

$C \wedge B$

This creates two sub-goals:

C

and

B

Step 2: Verify Cough

We have the fact:

C = True

Therefore:

C is proved

Step 3: Verify Breathlessness

We have the fact:

B = True

Therefore:

B is proved

Step 4: Satisfy the Original Goal

Since both:

C = True

and

B = True

are true, Rule 3 can be applied:

C ^ B → P

Therefore:

P = true

Goal Reduction Tree

```
                 Goal
        Patient A has Pneumonia (P)
                     |
                   Rule 3
                     |
              C AND B required
                 /        \
                /          \
        Cough (C)      Breathlessness (B)
            |                 |
        Given Fact         Given Fact
                \          /
                 \        /
                  \      /
              Pneumonia (P)
```

Backward Chaining Steps

P

↓ Apply Rule 3

C ^ B

↓ Verify facts

C = True, B = True

↓ Therefore

P = True

2. Ans: (a) Given Rule 1

Vehicle (x) ^ Emergency Type (x) → Clear Path (x)

Given Facts

Vehicle (Ambulance)

Emergency Type (Ambulance)

Step 1: Unify Vehicle(x) with Vehicle (Ambulance)

Compare:

Vehicle(x)

with:

Vehicle (Ambulance)

Therefore:

$\theta 1=\{x/Ambulance\}$

After substitution:

Vehicle (Ambulance)

Step 2: Unify Emergency Type(x) with Emergency Type (Ambulance)

Using the same substitution:

$\theta 2=\{x/Ambulance\}$

After substitution:

Emergency Type (Ambulance)

Both facts are available in the knowledge base.

Therefore, Rule 1 becomes:

Vehicle (Ambulance) ∧ Emergency Type (Ambulance) → Clear Path (Ambulance)

Hence:

Clear Path (Ambulance)

Most General Unifier (MGU)

$\theta=\{x/Ambulance\}$

(b) Given Rules

Rule 1:

Vehicle (x) ∧ Emergency Type (x) → Clear Path (x)

Rule 2:

Clear Path (x) → Green Signal (x) ∧¬Red Signal (x)

Rule 3:

Vehicle (x) ∧ Behind (x, y) ∧ Green Signal (y) → Proceed (x)

Step 1: Convert Rules into CNF

Rule 1

Vehicle (x) ∧ Emergency Type (x) → Clear Path (x)

Using:

$A \rightarrow B = \neg A \vee B$

and De Morgan's law:

¬Vehicle (x) ∨ ¬Emergency Type (x) ∨ Clear Path (x)

Rule 2

Clear Path (x) → Green Signal (x) ∧ ¬Red Signal (x)

Rule 3

Vehicle (x) ∧ Behind (x, y) ∧ Green Signal (y) → Proceed (x)

Step 2: Negate the Goal

Goal:

Proceed (CarA)

Negated goal:

¬Proceed (CarA)

Step 3: Derive Clear Path (Ambulance)

From Rule 1:

¬Vehicle (Ambulance) ∨ ¬Emergency Type (Ambulance) ∨ Clear Path (Ambulance)

Given:

Vehicle (Ambulance)

and:

Emergency Type (Ambulance)

Resolution gives:

Clear Path (Ambulance)

Step 4: Derive Green Signal (Ambulance)

From Rule 2:

¬Clear Path (Ambulance) ∨ Green Signal (Ambulance)

Using:

Clear Path (Ambulance)

we derive:

Green Signal (Ambulance)

Step 5: Derive Proceed (CarA)

Rule 3:

¬Vehicle (CarA) ∨ ¬Behind (CarA, Ambulance) ∨ ¬Green Signal (Ambulance) ∨ Proceed (CarA)

Step 6: Resolution with Negated Goal

We have:

Proceed (CarA)

and:

¬Proceed (CarA)

Resolution Tree

```
Vehicle (Ambulance)      Emergency Type (Ambulance)
          \                              /
           \                           /
            Clear Path (Ambulance)
                          |
                          |
             Green Signal (Ambulance)
                          |
                          |
Vehicle (CarA) + Behind (CarA, Ambulance)
                          |
                          ↓
              Proceed (CarA)
                         |
                         |  ¬Proceed (CarA)
                         ↓
                         □
```

(c) The current knowledge base can identify an emergency vehicle and derive a clear path for it using Rule 1:

Vehicle (x) ∧ Emergency Type (x) → Clear Path (x)

For example:

Emergency Type (Ambulance1)

and

Emergency Type (Ambulance2)

can result in:

Clear Path (Ambulance1)

and:

Clear Path (Ambulance2)

However, the current knowledge base is not sufficient for complete coordination of two simultaneous emergency vehicles.

It does not specify:

1. Which emergency vehicle should receive priority.
2. How conflicting routes should be handled.
3. What happens if both vehicles approach the same intersection.
4. How simultaneous green signals should be controlled.
5. How one emergency vehicle should wait for another.

3. Ans: (a) CNF Conversion

P1

Soil Dry → Irrigation Needed

Step 1 – Eliminate implication

Using:

A → B = ¬A ∨ B

P2

Irrigation Needed ∧ Crop Wheat → Apply Drip Method

Eliminate implication:

¬(Irrigation Needed ∧ Crop Wheat) ∨ Apply Drip Method

Using De Morgan's law:

¬Irrigation Needed ∨ ¬Crop Wheat ∨ Apply Drip Method

P3

Eliminate implication:

¬Apply Drip Method ∧ Crop Wheat → Crop at Risk

Apply De Morgan's law:

¬ (¬Apply Drip Method ∧ Crop Wheat) ∨ Crop at Risk

(b) Resolution Refutation for Apply Drip Method

Goal

Apply Drip Method

For resolution refutation, negate the goal:

¬Apply Drip Method

Add this to the KB.

Step 1

We have:

Soil Dry

and:

¬Soil Dry ∨ Irrigation Needed

Resolve them:

Irrigation Needed

Step 2

We have:

Irrigation Needed

and:

Crop Wheat

and:

¬Irrigation Needed ∨ ¬Crop Wheat ∨ Apply Drip Method

Step 3 – Contradiction

We added:

¬Apply Drip Method

But resolution has derived:

¬Apply Drip Method

Therefore:

Apply Drip Method + ¬Apply Drip Method

Resolution Proof Tree

```
      SoilDry
        |
        |  C1
        ↓
   Irrigation Needed
        |
```

| + Crop Wheat

↓

Apply Drip Method

|

| + ¬Apply Drip Method

↓

□

(c) Remove SoilDry

Initially:

Soil Dry = True

was responsible for deriving:

Irrigation Needed

If SoilDry is removed, Rule P1:

Soil Dry → Irrigation Needed

cannot be activated.

Therefore:

Irrigation Needed

cannot be derived.

Since P2 requires:

Irrigation Needed ∧ Crop Wheat

P3 is:

¬Apply Drip Method ∧ Crop Wheat → Crop at Risk

We know:

Crop Wheat = True

Final Analysis

Fact/Conclusion	Before removing SoilDry	After removing SoilDry
Soil Dry	True	Not available
Crop Wheat	True	True
Irrigation Needed	Derived	Not derived
Apply Drip Method	Derived	Not derived
¬Apply Drip Method	Not given	Not given
Crop at Risk	Not derived	Not derived

4. (a) Belief State and Modus Ponens

Initial percepts

The agent perceives:

Stench [1,2]

Breeze [1,1]

Glitter [2,2]

Apply Rule 1

Stench [1,2] → Wumpus Adjacent [1,2]

Since:

Stench [1,2] = True

Therefore, a Wumpus may be in a cell adjacent to [1,2].

Apply Rule 2

Since:

Breeze [1,1] = True

Apply Rule 3

Given:

Glitter [2,2]

we derive:

Glitter [2,2]

Therefore, the gold is at [2,2].

Percepts:

- Stench [1,2]
- Breeze [1,1]
- Glitter [2,2]

Derived:

- Wumpus Adjacent [1,2]
- Pit Adjacent [1,1]
- Gold [2,2]

(b) Forward Chaining – Possible Wumpus and Pit Locations

Wumpus

Stench [1,2]

From:

Wumpus Adjacent [1,2]

we derive:

Therefore, possible Wumpus locations are:

[1,1], [1,3], [2,2]

Reasoning Table

Percept	Rule	Derived knowledge	Possible locations
Stench [1,2]	Rule 1	Wumpus adjacent to [1,2]	[1,1], [1,3], [2,2]
Breeze [1,1]	Rule 2	Pit adjacent to [1,1]	[1,2], [2,1]
Glitter [2,2]	Rule 3	Gold [2,2]	[2,2]

Important Logical Observation

The rules tell us that a Wumpus or Pit is adjacent, not exactly which adjacent cell contains it. Therefore, selecting one particular cell as the Wumpus or Pit would be an unsupported conclusion.

(c) Safety of Path [1,1] → [1,2] → [2,2]

The proposed path is:

[1,1] → [1,2] → [2,2]

We must determine whether every cell can be proven safe.

Cell [1,1]

We know:

Breeze [1,1]

This means a Pit is adjacent to [1,1].

Possible Pit locations include:

[1,2], [2,1]

Therefore, [1,1] itself is not proven to contain a Pit, but it is also not proven safe by Rule 4.

Cell [1,2]

We know:

Stench [1,2]

This indicates that a Wumpus is adjacent to [1,2].

Possible Wumpus locations include:

[1,1], [1,3], [2,2]

Thus, [1,2] itself is not directly identified as containing the Wumpus.

However, it is not proven safe either.

Cell [2,2]

We know:

Glitter [2,2]

Therefore:

Gold [2,2]

The agent wants to reach this cell to collect the gold.

But [2,2] is also one of the possible Wumpus locations derived from the Stench at [1,2]:

[2,2] $\in$ Possible Wumpus Locations

Therefore, entering [2,2] has a potential hazard.